Engine Model(s): PT6A

71-00-00

POWER PLANT - ADJUSTMENT/TEST

1. General

This section contains information for engine ground running and for checks and adjustment/test procedures to verify the integrity of installed/replaced items. Before making any control adjustments/checks, the engine should be run for ten minutes minimum and all parameters stabilized.

NOTE: Although generalized procedures for ground testing are described in the following paragraphs, the appropriate Aircraft Manufacturer's Manual must be consulted for detailed procedures, reference curves and/or tables required to do the tests.

A. Specific checks and the order in which they are to be done after installation or replacement of components are defined in Table 501.

Table 501 Checks after Component Installation/Replacement

Item Installed/Replaced	Check Req'd (Para. No.)
Compressor Bleed Valve	4.7.13.8. 9.
Engine on Airframe	3.5.6.4.7.8.9.11.A.14.11.C.10.13.
Scavenge Oil Tubes	4.7.8.9.
Starting Flow Control Unit/Flow Divider	6.5.4.7.8.9.
Fuel Control Unit	6.5.4.7.11.A.14.10.8.9.
Fuel Drain Valves (Front and Rear)	6.5.4.7.8.9.
Fuel Lines	6.5.4.7.8.9.
Fuel Manifold/Fuel Nozzles	6.5.4.7.8.9.13.
Fuel Pump	6.5.4.7.8.9.
Shaft Oil Seals	4.7.8.9.
Oil Filter Housing and Check Valve	5.6.4.7.8.9.
Oil Pressure Relief Valve	4.7.11.C.8.
Oil-to-Fuel Heater	6.5.4.7.8.9.
Pressure Oil System	4.7.11.C.8.9.
Propeller Governor	4.7.14.10.8.9.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Item Installed/Replaced	Check Req'd (Para. No.)
P3 (Compressor) Air Filter (when fitted)	4.7.8.9.
Reversing Linkage	4.7.15.8.9.

B. Minimum checks to be done before fault isolation procedures are initiated and done on an engine are defined in Table 502.

Table 502 Minimum Checks Required before Troubleshooting

Check and Sequence	Para. No.
Engine Pre-operational Check	3.
Engine Pre-start Check	4.
Dry Motoring Run	5.
Wet Motoring Run	6.
Engine Start	7.
Engine Shutdown	8.

C. Engine operating limits are defined in Tables 503, 504 and 505.

Table 503 PT6A-21 Engine Operating Limits

Power Setting (2)	SHP	Torque (1)		Max. Observed ITT °C	Ng (9)		Np		Oil Press. psig (3)	Oil Temp. °C (4)
		lb. ft.	psig		RPM	%	RPM	%		
Takeoff and Max. Cont.	550 32.8°C (91°F)	1315	42.5	695	38100	101.5	2200	100	80 to 100	10 to 99°C (50 to 210°F)
Max. Climb/ Max.Cruise	550 47.2°C (117°F)	1315	42.5	695	38100		2200	100	80 to 100	0 to 99°C (32 to 210°F)
Idle				660 (5)					40 Min.	-40 to 99 °C (-40 to 210 °F)

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Power Setting (2)	**SHP**	**Torque (1)**		**Max. Observed ITT °C**	**Ng (9)**		**Np**		**Oil Press. psig (3)**	**Oil Temp. °C (4)**
		lb. ft.	**psig**		**RPM**	**%**	**RPM**	**%**		
Starting				1090 (6)(8)						-40 Min.
Acceleration		1500 (6)	48.5 (6)	825 (6)	38500	102.6	2420	110		0 to 99 °C (32 to 210 °F)
Max. Reverse	500 32.8°C (91°F)	1315 (7)	42.5 (7)	695	38100	101.5	2112 2068	96 94	80 to 100	0 to 99 °C (32 to 210 °F)

NOTE: 1. Maximum permissible sustained torque is 1315 lb.ft. (42.5 psig). Np must be set so as not to exceed power limitations.

NOTE: 2. For every 10°C (18°F) below -30°C (-22°F) ambient temperature, reduce maximum allowable Ng by 2.2%.

NOTE: 3. Minimum oil pressure above 28000 Ng is 80 psig.

NOTE: 4. For increased service life of engine oil, and oil temperature below 80°C (176°F) is recommend. A minimum oil temperature of of 55 °C (130 °F) is recommended for fuel heater operation at take-off power.

NOTE: 5. At 19000 rpm (Ng) minimum. Advance power lever to maintain this value.

NOTE: 6. These values are time limited to 2 seconds.

NOTE: 7. Reverse power operation is limited to one minute.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Power Setting (2)	SHP	Torque (1)		Max. Observed ITT °C	Ng (9)		Np		Oil Press. psig (3)	Oil Temp. °C (4)
		lb. ft.	psig		RPM	%	RPM	%		
NOTE: 8. Starting temperatures above 925 °C should be investigated for cause.										
NOTE: 9. All limits are based on sea level and ambient temperatures as specified.										

Table 504 PT6A-27 Engine Operating Limits

Power Setting (2)	SHP	Torque (1)		Max. Observed ITT °C	Ng (9)		Np		Oil Press. psig (3)	Oil Temp. °C (4)
		lb. ft.	psig		RPM	%	RPM	%		
Takeoff and Max. Cont.	680 21.7°C (71°F)	1628	53.3	725	38100	101.5	2205	100	80 to 100	10 to 99°C (50 to 210°F)
Max. Climb/ Max.Cruise	620 20.6°C (69°F)	1628	53.3	695	38100	101.5	2205	100	80 to 100	0 to 99°C (32 to 210°F)
Idle				660 (5)					40 Min.	-40 to 99 °C (-40 to 210 °F)
Starting				1090 (6)(8)						-40 Min.
Acceleration		2100 (6)	68.7 (6)	825 (6)	38500 (6)	102.6	2420 (10)	110		0 to 99 °C (32 to 210 °F)
Max. Reverse	620	1628 (7)	53.3 (7)	725	38100	101.5	2112 2068	96 94	80 to 100	0 to 99 °C (32 to 210 °F)

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Power Setting (2)	SHP	Torque (1)		Max. Observed ITT °C	Ng (9)		Np		Oil Press. psig (3)	Oil Temp. °C (4)
		lb. ft.	psig		RPM	%	RPM	%		
NOTE: 1. Maximum permissible sustained torque is 1628 lb.ft. (53.3 psig). Np must be set so as not to exceed power limitations.										
NOTE: 2. For every 10°C (18°F) below -30°C (-22°F) ambient temperature, reduce maximum allowable Ng by 2.2%.										
NOTE: 3. Minimum oil pressure above 28000 Ng is 80 psig.										
NOTE: 4. For increased service life of engine oil, and oil temperature below 80°C (176°F) is recommend. A minimum oil temperature of of 55 °C (130 °F) is recommended for fuel heater operation at take-off power.										
NOTE: 5. At 19000 rpm (Ng) minimum. Advance power lever to maintain this value.										
NOTE: 6. These values are time limited to 2 seconds.										
NOTE: 7. Reverse power operation is limited to one minute.										
NOTE: 8. Starting temperatures above 925 °C should be investigated for cause.										
NOTE: 9. All limits are based on sea level and ambient temperatures as specified.										
NOTE: 10. In the event of failure of the propeller governor toward overspeed, it is permissible to complete a flight with propeller control via the overspeed governor (on engines so equipped) providing this limit is not exceeded.										

Table 505 PT6A-28 Engine Operating Limits

Power Setting (2)	SHP	Torque (1)		Max. Observed ITT °C	Ng (9)		Np		Oil Press. psig (3)	Oil Temp. °C (4)
		lb. ft.	psig		RPM	%	RPM	%		
Takeoff and Max. Cont.	680 21.7 °C (71 °F)	1628	53.3	750	38100	101.5	2205	100	80 to 100	10 to 99°C (50 to 210°F)

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Power Setting (2)	SHP	Torque (1)		Max. Observed ITT °C	Ng (9)		Np		Oil Press. psig (3)	Oil Temp. °C (4)
		lb. ft.	psig		RPM	%	RPM	%		
Norm. Cruise	620 20.6 °C (69 °F)	1628	53.5	695	38100		2205	100	80 to 100	0 to 99 (32 to 210 °F)
Max. Climb/ Max.Cruise	620 32.8 °C (91 °F)	1628	53.3	750	38100	101.5	2205	100	80 to 100	0 to 99°C (32 to 210°F)
Idle				660 (5)					40 Min.	-40 to 99 °C (-40 to 210 °F)
Starting				1090 (6)(8)						-40 Min.
Acceleration		2100 (6)	68.7 (6)	850 (6)	38500 (6)	102.6	2420 (10)	110		0 to 99 °C (32 to 210 °F)
Max. Reverse	620	1628 (7)	53.3 (7)	750	38100	101.5	2090	95	80 to 100	0 to 99 °C (32 to 210 °F)

NOTE: 1. Maximum permissible sustained torque is 1628 lb.ft. (53.3 psig). Np must be set so as not to exceed power limitations.

NOTE: 2. For every 10°C (18°F) below -30°C (-22°F) ambient temperature, reduce maximum allowable Ng by 2.2%.

NOTE: 3. Minimum oil pressure above 28000 Ng is 80 psig.

NOTE: 4. For increased service life of engine oil, and oil temperature below 80°C

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

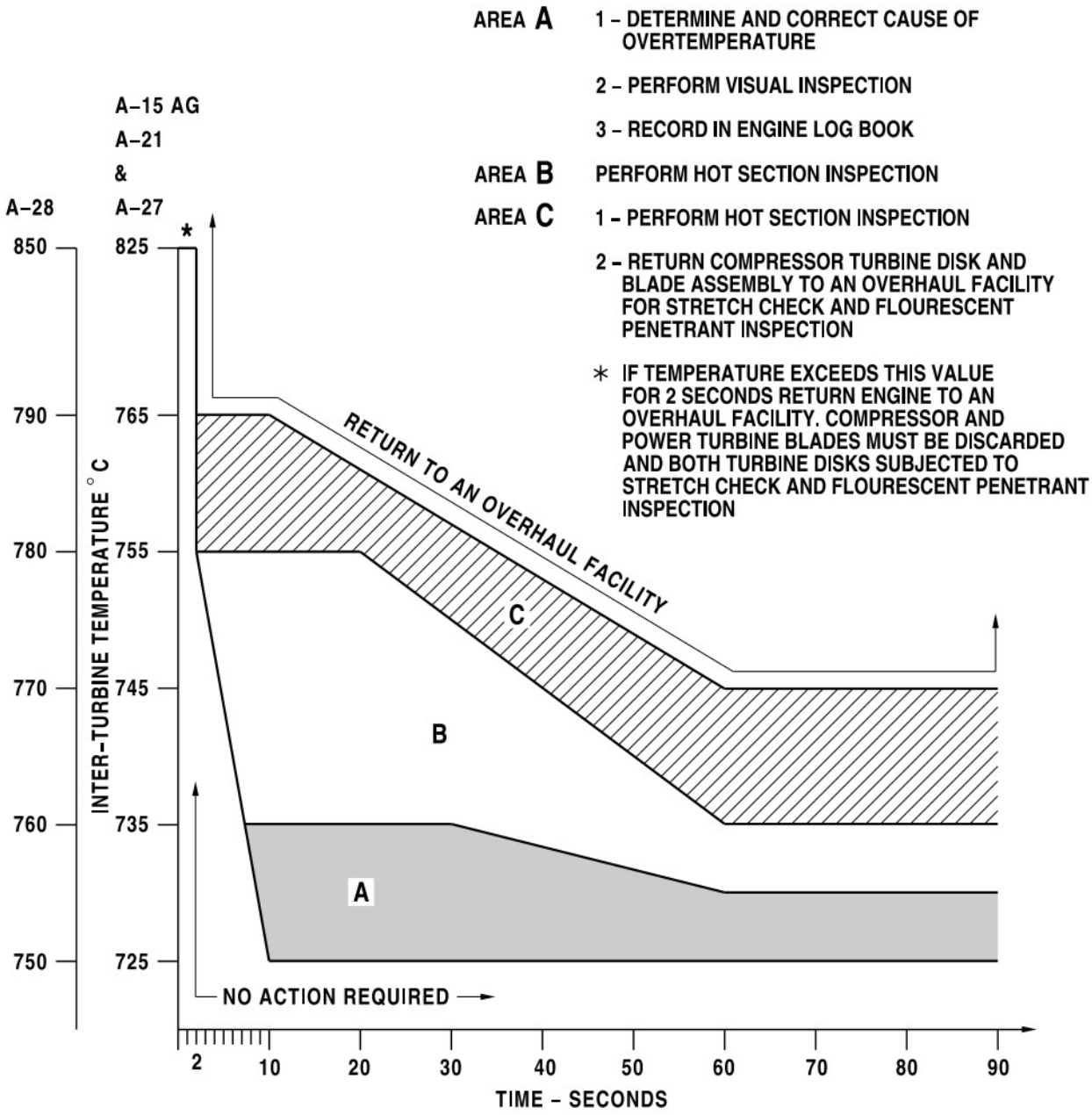
Power Setting (2)	SHP	Torque (1)		Max. Observed ITT °C	Ng (9)		Np		Oil Press. psig (3)	Oil Temp. °C (4)
		lb. ft.	psig		RPM	%	RPM	%		
(176°F) is recommend. A minimum oil temperature of of 55 °C (130 °F) is recommended for fuel heater operation at take-off power.										
NOTE: 5. At 19000 rpm (Ng) minimum. Advance power lever to maintain this value.										
NOTE: 6. These values are time limited to 2 seconds.										
NOTE: 7. Reverse power operation is limited to one minute.										
NOTE: 8. Starting temperatures above 925 °C should be investigated for cause.										
NOTE: 9. All limits are based on sea level and ambient temperatures as specified.										
NOTE: 10. In the event of failure of the propeller governor toward overspeed, it is permissible to complete a flight with propeller control via the overspeed governor (on engines so equipped) providing this limit is not exceeded.										

2. Engine Overtorque and Overtemperature Limits

Refer to Figure 501 and Figure 502 for engine overtemperature limits and corresponding inspection requirements.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

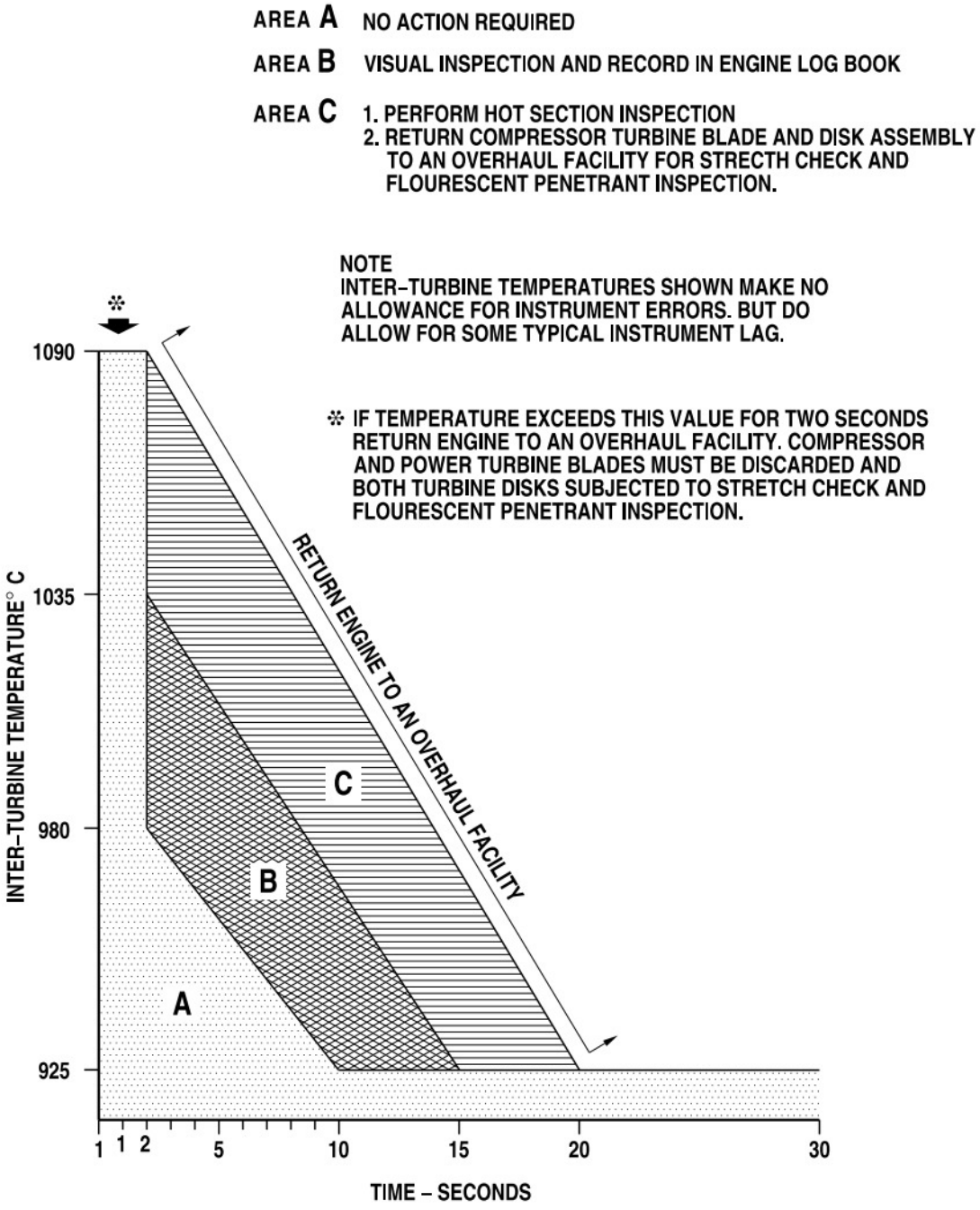
Figure 501 Overtemperature Limits (All Conditions Except Starting)



c467h

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

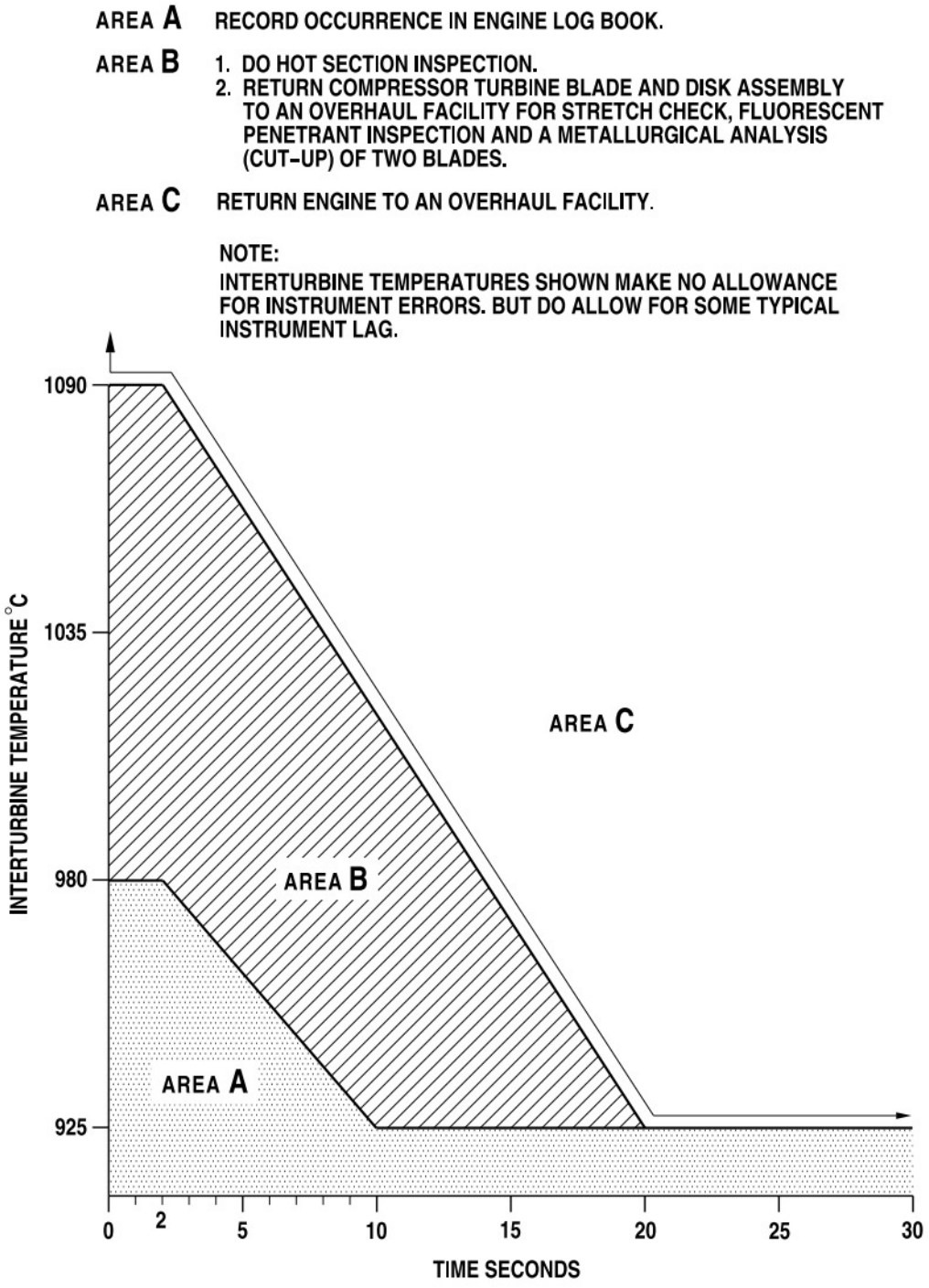
Figure 502 Overtemperature Limits (Starting Conditions Only)



c233f

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 503 Inadvertent Cut-off and Relight During Taxi



c94193

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

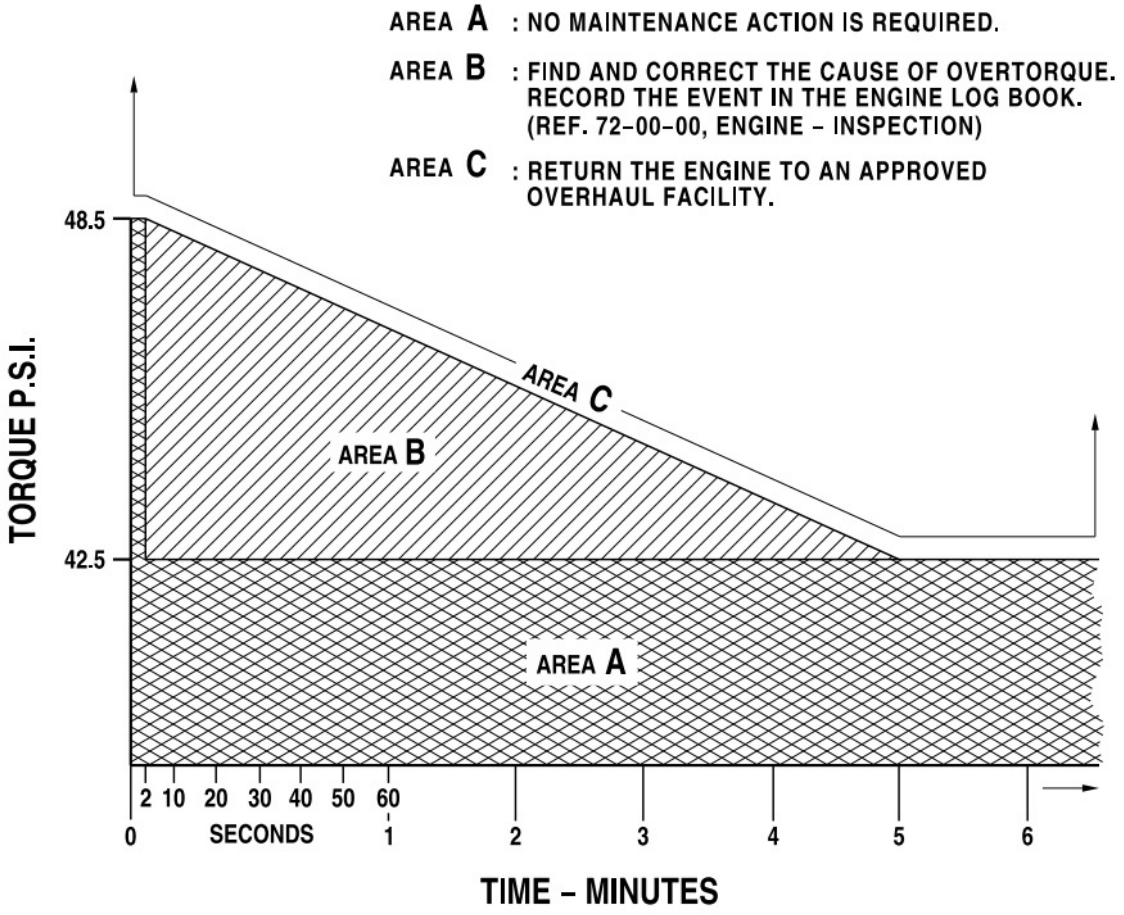
Refer to Figure 503 for engine inadvertent cut-off and relight during taxi and corresponding inspection requirements. For unscheduled inspection procedure, refer to Chap. 72-00-00, INSPECTION, Unscheduled Inspection.

Refer to Figure 504 for engine overtorque limits and corresponding maintenance actions.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 504 Overtorque Limits (All Conditions)

(SHEET 1 OF 2)

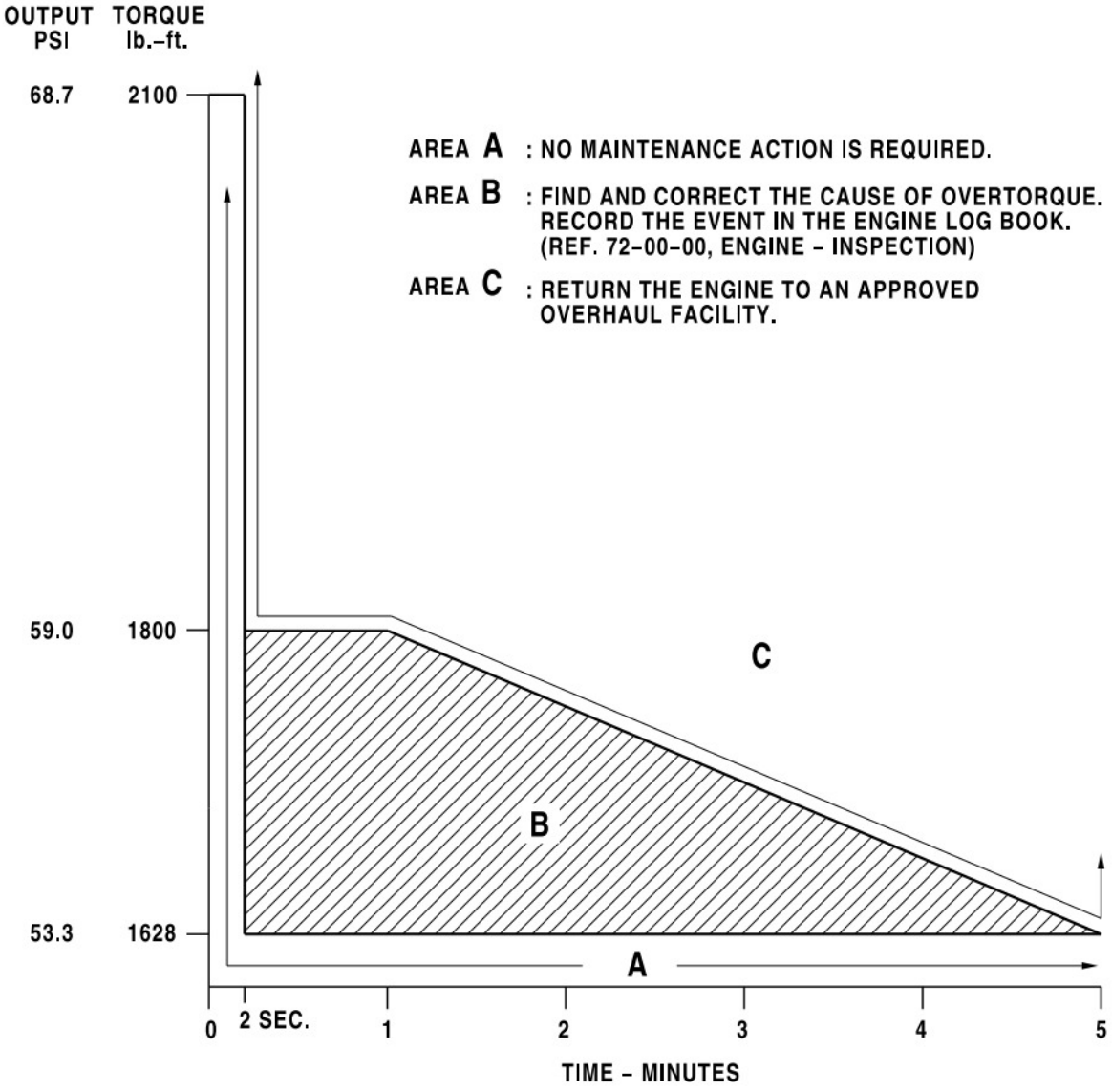


PT6A-21

c192514

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

(SHEET 2 OF 2)



PT6A-27 and PT6A-28

c192515

3. Engine Pre-operational Check

A. Procedure

(1) Remove the air inlet and exhaust covers (if fitted). Do not permit the propeller to windmill if a restraining device is secured to the exhaust covers.

(2) Open engine cowlings (Ref. Aircraft Manufacturer's Manual) and inspect the engine mounts and air inlet screen, and spray ring, if fitted.

(3) Examine all linkages to the engine and to the accessories mounted on the engine, for attachment.

(4) Examine all fuel and oil tubes, lines and connections for security and for leaks. No leaks are permitted except at normal seepage drains.

(5) Examine all pneumatic line connections and filters for security.

WARNING: MAKE SURE THE ELECTRICAL POWER IS OFF.

(6) Examine all electrical connectors for proper installation.

CAUTION: DO NOT MOVE THE COCKPIT POWER CONTROL LEVER INTO PROPELLER REVERSE POSITION, AS DAMAGE TO THE LINKAGE WILL RESULT. REVERSE MAY ONLY BE SELECTED WITH THE ENGINE RUNNING AND THE PROPELLER TURNING.

(7) Check the cockpit engine controls for freedom of movement and verify full travel of engine control linkages.

NOTE: Individual airframe installations may prevent full application of this check with the electrical power off and engine not running. Refer to individual aircraft operating instructions.

(8) Check engine oil level and that oil system has been properly serviced (Ref. 72-00-00, SERVICING). Check locking of the filler cap after servicing.

(9) Close engine cowlings (Ref. Aircraft Manufacturer's Manual).

(10) Make sure that fuel system has been serviced with proper fuel (Ref. Service Bulletin SB1244, Approved Listing of Engine Fuel and Additives).

(11) If the engine is not to be started immediately, refit the air inlet and exhaust covers. If the engine is to be started, remove the propeller restraining device. Check that the area is clear for engine motoring.

4. Pre-start Checks

A. Engine Control and Switch Position

NOTE: Cockpit labels and names of controls tend to differ according to the airframe manufacturer. The terminology used in this and subsequent paragraphs is that of Pratt & Whitney Canada.

(1) Power Control Lever - GROUND IDLE.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

(2) Propeller Control Lever - Anywhere in operating range.

(3) Fuel Condition Lever - OFF (PT6A-21); Starting Control Lever - CUT-OFF (PT6A-27 and PT6A-28).

WARNING: MAKE SURE PERSONNEL, EQUIPMENT AND OTHER AIRCRAFT ARE CLEAR OF DANGER ZONE NEAR ENGINE.

(4) If proceeding to a start:

(a) Engine Master Switch - ON.

(b) Fuel System Shut-off Valve - OPEN.

(c) Fuel Boost Pump Switch - ON.

(d) Fuel Inlet Pressure Indicator - Check 5 psig minimum .

(e) Bleed Air - OFF.

5. Dry Motoring Run

A. Procedure

NOTE: This procedure is used at any time it is deemed necessary to remove internally trapped fuel and vapor, or if there is evidence of a fire within the engine gas path. Air passing through the engine serves to purge fuel, vapor or fire from the combustion section, compressor turbine, power turbines, and exhaust system.

(1) Fuel Condition Lever - OFF (PT6A-21); Starting Control Lever - CUTOFF (PT6A-27 and PT6A-28).

(2) Ignition Switch - OFF.

(3) Engine Master Switch - ON.

(4) Fuel System Shut-off Valve - OPEN.

(5) Fuel Boost Pump Switch - ON.

(6) Engine Starter Switch - ON.

WARNING: SHOULD THE FIRE PERSIST, AS INDICATED BY SUSTAINED T5, CLOSE THE FUEL SYSTEM SHUT-OFF VALVE AT THIS TIME AND CONTINUE MOTORING.

(7) Maintain the starter ON for ten (10) seconds, observing appropriate starter limits (Ref. Starter Manufacturer's Manual).

(8) Engine Starter Switch - OFF

(9) Fuel Boost Pump Switch - OFF.

(10) Fuel System Shut-off Valve - CLOSED.

(11) Engine Master Switch - OFF.

(12) Check all oil and fuel lines and connections for leaks. None are allowed.

(13) Check engine oil level and service as necessary (Ref. 72-00-00, SERVICING).

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

6. Wet Motoring Run

CAUTION: THIS PROCEDURE IS USED STRICTLY FOR MAINTENANCE PURPOSES AND IS NOT PART OF THE NORMAL START PROCEDURE. AFTER A WET MOTORING RUN, A DRY MOTORING RUN (REF. PARA. 5.) MUST BE ACCOMPLISHED BEFORE ANY START IS ATTEMPTED.

A. Procedure

(1) Fuel Condition Lever - OFF.

(2) Ignition Switch - OFF.

(3) Engine Master Switch - ON.

(4) Fuel System Shut-off Valve - OPEN.

(5) Fuel Boost Pump Switch - ON.

(6) Engine Starter Switch - ON.

(7) After the gas generator has reached stabilized speed, Fuel Condition Lever to GROUND-IDLE for 10 seconds.

(8) Fuel Condition Lever - OFF (PT6A-21); Starting Control Lever - CUTOFF (PT6A-27 and PT6A-28).

(9) Engine Starter Switch - OFF.

(10) During rundown, check that both gas generator case drain valves are draining.

(11) Fuel Boost Pump Switch - OFF.

(12) Fuel System Shut-off Switch - CLOSED.

(13) Engine Master Switch - OFF.

(14) Check all engine fuel lines and connections for leaks. None are allowed.

7. Engine Starting

A. Procedure

(1) Engine Starter Switch - ON.

(2) Engine Oil Pressure - check for indication.

(3) Ignition Switch - ON.

NOTE: The minimum gas generator rpm to obtain a satisfactory light is 4500 rpm (12% Ng).

(4) For engines equipped with glow-plug ignition, after the gas generator speed passes through, or has stabilized above, the minimum light-off speed, wait for an additional five seconds (approximately) before advancing the Starting Control (Fuel Condition) lever to GROUND-IDLE.

(5) For engines equipped with spark ignition, after the gas generator speed passes through or has stabilized above the minimum light-off speed, advance Fuel Condition or Starting Control Lever to GROUND-IDLE.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

(6) Make sure that engine accelerates normally to GROUND-IDLE rpm and that maximum allowable ITT starting limit is not exceeded.

CAUTION: WHENEVER GAS GENERATOR FAILS TO LIGHT UP WITHIN 10 SECONDS AFTER MOVING FUEL CONDITION OR STARTING CONTROL LEVER TO GROUND-IDLE POSITION, SHUT OFF FUEL, STARTER, AND IGNITION. ALLOW A 30 SECOND FUEL DRAINING PERIOD, FOLLOWED BY DRY MOTORING RUN. REPEAT COMPLETE STARTING SEQUENCE; OBSERVE STARTER OPERATING LIMITS.

(7) When Ground-Idle is reached:

(a) Starter - OFF.

(b) Ignition - OFF.

(8) Do two feathering cycles to bleed propeller governor and propeller hub.

8. Engine Shutdown

A. Procedure

WARNING: IF THERE IS ANY EVIDENCE OF FIRE WITHIN THE ENGINE AFTER SHUTDOWN, PROCEED IMMEDIATELY AS DESCRIBED UNDER DRY MOTORING RUN (REF. PARA. 5.).

(1) Power Control Lever - GROUND-IDLE.

NOTE: Allow engine to stabilize for a minimum of one minute at minimum obtainable inter-turbine temperature.

(2) Propeller Control Lever - FEATHER.

(3) Fuel Condition Lever - OFF (PT6A-21); Starting Control Lever - CUTOFF (PT6A-27 and PT6A-28).

(4) Fuel Boost Pump Switch - OFF.

(5) Fuel System Shut-off Switch - CLOSED.

(6) Engine Master Switch - OFF.

NOTE: Make sure that the compressor decelerates freely. The compressor rundown time must not be less than 20 seconds from ground idle.

9. Post-Shutdown Checks

A. Procedure

(1) Check all engine oil lines and connections for leaks. None are allowed, except at normal seepage drains as specified in Table 506.

Table 506 Engine Oil and Fuel Permissible Leakage

LOCATION OF DRAIN	MAXIMUM LEAKAGE DURING OPERATION
Engine Oil Breather	15 cc/hr

LOCATION OF DRAIN	MAXIMUM LEAKAGE DURING OPERATION
Fuel Control Unit	Zero
Fuel Pump	0.55 cc/min (seepage)
Gas Generator Drain Valves	Zero
Oil Consumption	0.2 lb/hr (100 cc/hr) or 1 liter in 10 hours
Oil Filter Housing Check Valve	0.50 cc/hr
Start Control or Dump Valve (as applicable)	0.2 cc/min
Starter Oil Seal Cavity	3.0 cc/hr
NOTE: When a specific seal leakage is not specified, the seal leakage shall not exceed 3 cc per hour per individual seal. All seal leakages shall be included in total oil consumption (Ref. Chapter 72-00-00, Engine Description and Operation).	

(2) Check engine oil level and service as necessary (Ref. Chapter 72-00-00, SERVICING).

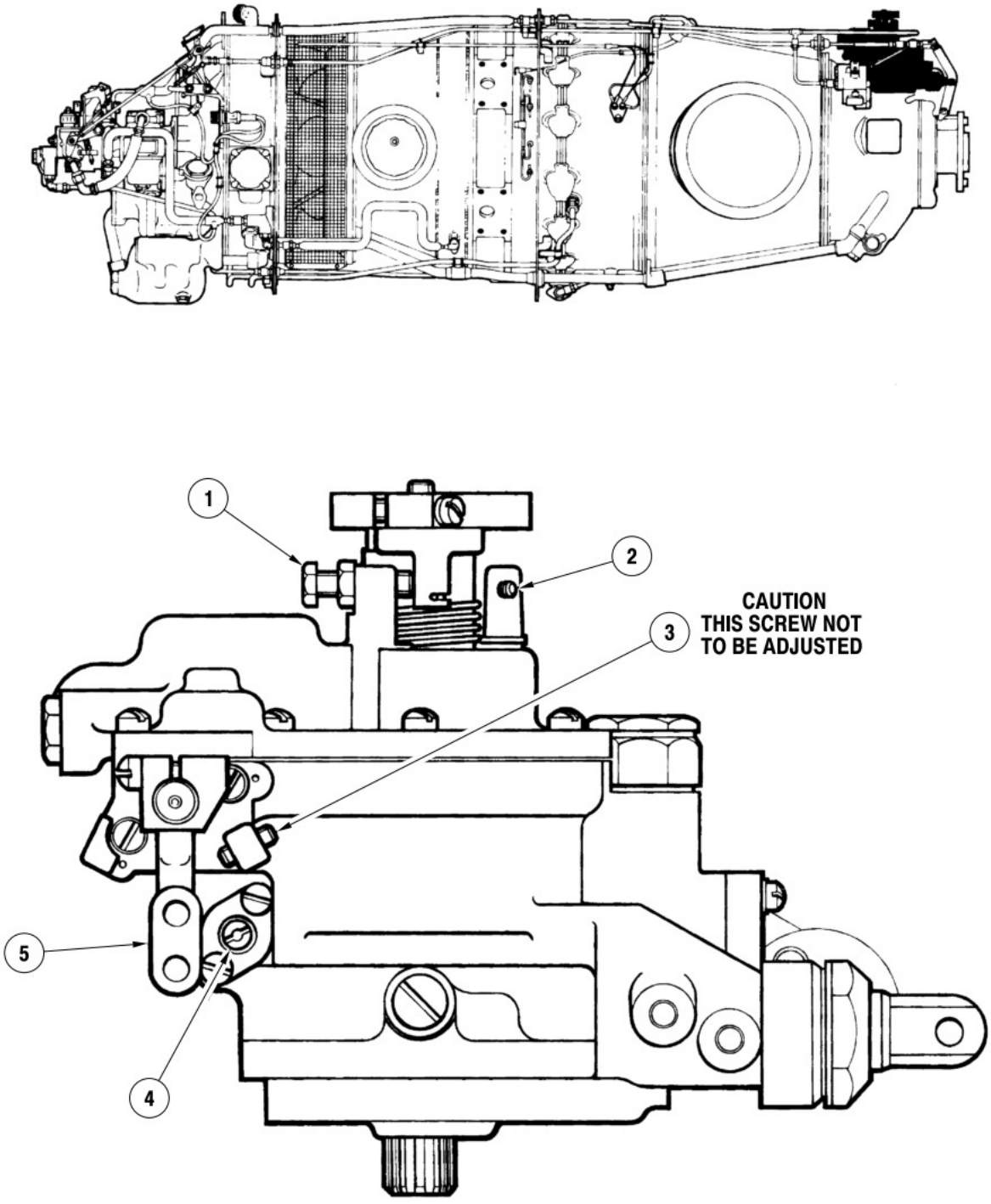
(3) Check all engine-mounted accessories and linkages for security for mounting.

(4) Check oil-to-fuel heater installation (Ref. 73-10-01, MAINTENANCE PRACTICES).

10. Propeller Governor Checks (Ref. Fig. 505)

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 505 Propeller Governor Adjustments



c930

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

1. Governor Maximum Speed Adjuster
2. Feathering Adjuster
3. Pneumatic Maximum Adjuster (Not for Field Use)
4. Pneumatic Minimum Adjuster
5. Airbleed Link (Reset Arm) - Pneumatic Section

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

A. Procedure

CAUTION: MAKE SURE THAT THE Py TUBE IS BLANKED. ALSO BLANK THE FITTING ON THE GOVERNOR TO STOP OIL FROM SEEPING OUT DURING ENGINE RUN.

(1) With the engine shut down, disconnect and cap the governor Py air pressure tube at the governor (Ref. 73-10-08).

(2) Do the pre-start checks (Ref. Para. 4.).

(3) Start the engine and operate for five minutes at ground idle to permit warm-up of the engine oil to 75°C (167°F) minimum.

(4) Advance the power control lever to obtain 80% Ng, and perform two feathering cycles to purge air from the system and propeller hub.

NOTE: Reduce feathering time by turning the feathering adjuster (2) in a counterclockwise direction. Two turns from the nominal is the maximum permissible adjustment.

(5) Set the propeller control lever to the full fine pitch position. Advance the power control lever sufficiently for the propeller to reach constant speed, and mark the position on the cockpit quadrant. Note and record the Ng.

(6) Adjust the governor maximum speed adjuster (1) as required to get the necessary propeller speed (Np). An adjustment to counterclockwise increases the speed.

NOTE: Maximum Np can vary with different installations and according to the model. Refer Aircraft Manufacturer's Manual for the correct setting.

(7) Reconnect the governor Py air pressure tube to governor (Ref. 73-10-08).

(8) Set the propeller control to a full fine pitch position, and the power control lever to the quadrant position marked in step (5) .

(9) Note the Ng and compare it with value recorded in step (5) . The maximum permissible Ng drop is 100 rpm. If exceeded, check the linkage to make sure the reset arm is in contact with the maximum stop. Adjust if necessary.

CAUTION: DO NOT DISTURB THE PNEUMATIC MAXIMUM ADJUSTER (3) UNDER ANY CONDITIONS.

(10) Repeat steps (8) and (9) . If not satisfactory, change the propeller governor.

(11) Disconnect linkage attached to the airbleed link (5) and attach the airbleed link (5) to the minimum rear position.

(12) With the propeller control lever in fine pitch, move power control lever to the maximum stop and record the propeller shaft speed (Np). Make sure that the Np is 5 ± 1% less than the speed recorded in Step (6). Adjust the pneumatic minimum adjustment (4) in increments to get the necessary propeller shaft speed (Np).

11. Fuel Control Unit Checks

A. Minimum Governing

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

NOTE: Adjustments must be done under conditions representative of normal operation as instructed in relevant Aircraft Maintenance Manual (i.e., specified loads).

CAUTION: DO NOT PERFORM CHECKS AT BAROMETRIC PRESSURES BELOW 25.2 IN. Hg OR ALTITUDES IN EXCESS OF 3500 FT. CHECKS MUST BE DONE WITH ZERO POWER EXTRACTION.

(1) General

(a) At barometric pressures equivalent to altitudes less than 3500 feet above sea level, the fuel flow is metered to maintain ground-idle speed at a constant value, and is referred to as the minimum governing speed. However, since the fuel flow required to maintain a constant ground-idle speed decreases with increasing altitude, a point is reached where the fuel flow does not decrease further, this point being the minimum fuel flow setting. At altitudes higher than this point, the gas generator speed increases proportionally with altitude.

(b) Usually, the minimum fuel flow setting will be lower than the minimum governing speed. However, under marginal conditions, that is, when the altitude is close to or slightly higher than 3500 feet and/or barometric pressure is close to or slightly less than 25.2 in. Hg, the two speeds may coincide, or the Ng at minimum fuel flow may be higher than that at minimum governing.

(c) From the preceding, it will be seen that ground-idle speed adjustments should only be done at conditions of low altitude and/or high barometric pressure.

CAUTION: IF INTERTURBINE TEMPERATURE (T5) RISES ABNORMALLY AT ANY TIME DURING GROUND-IDLE SPEED, SHUT DOWN ENGINE IMMEDIATELY.

(2) Start engine (Ref. Para. 7.).

(3) Fuel Condition Lever - GROUND-IDLE (PT6A-21); Starting Control Lever - RUN (PT6A-27 and PT6A-28).

(4) Power Control Lever - GROUND-IDLE.

(5) Check that Ng is 52.0%. If not:

(a) Shut down engine (Ref. Para. 8.).

(b) Remove lockwire on adjuster (2, Fig. 506 or 507, as appropriate).

(c) Prevent adjuster screw from turning, and loosen nut.

(d) Turn adjuster (2) in increments until Ng equals 52.0%.

NOTE: To increase speed, turn adjuster clockwise.

(6) Restart engine and repeat steps (3) through (5). If Ng remains offspeed repeat foregoing procedure until required speed is obtained.

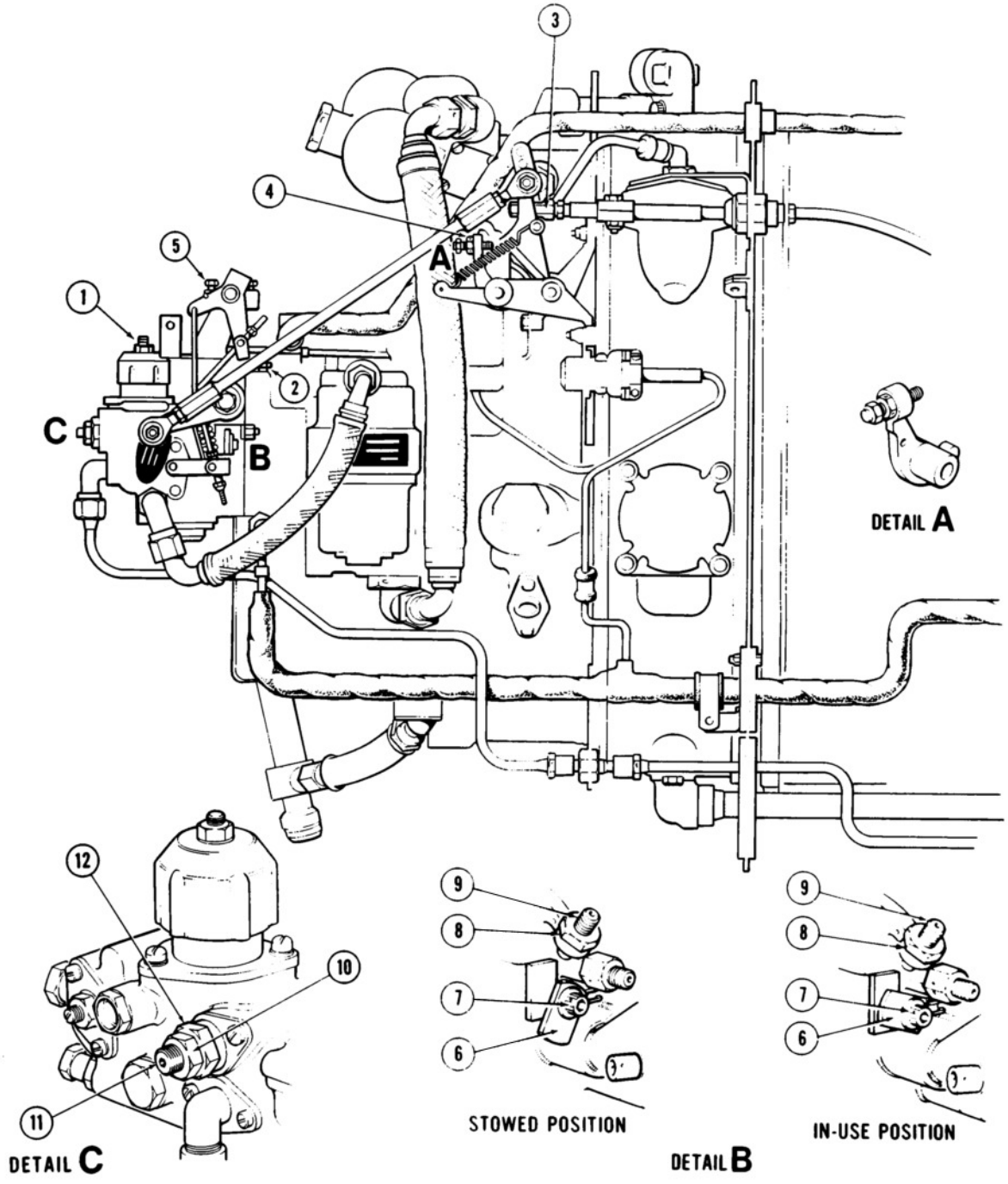
(7) Tighten jam nut to lock adjuster (2), torque 20 to 25 lb.in. and secure adjuster with lockwire.

(8) Shut down engine (Ref. Para. 8.).

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

B. FLIGHT-IDLE Adjustment

Figure 506 Fuel Control Unit Adjustments



PT6A-21

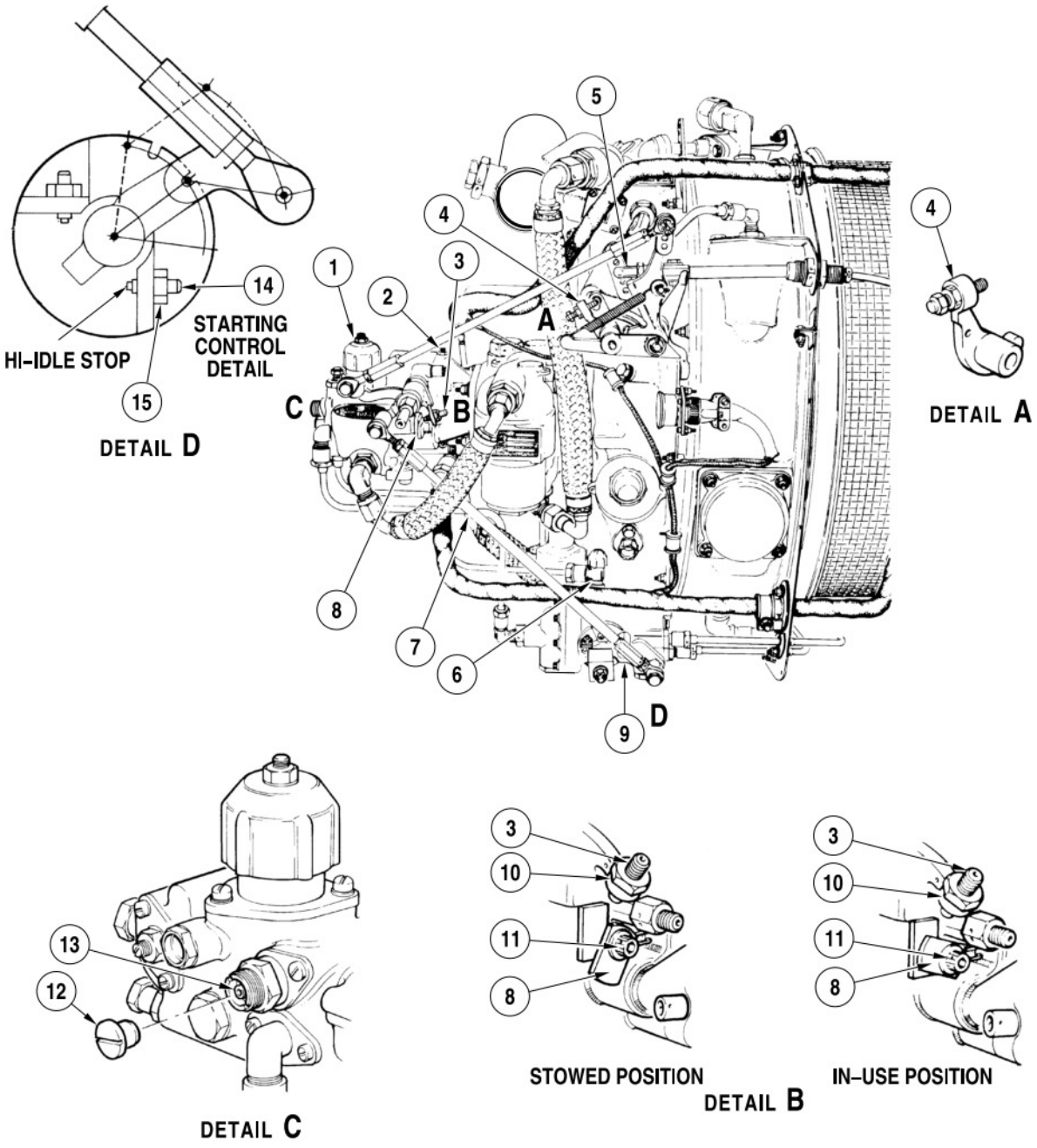
c3821b

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

1. Acceleration Adjustment Dome
2. Minimum Governing Speed Adjuster
3. Rear End Clevis
4. GROUND-IDLE Deadband Adjuster
5. FLIGHT-IDLE Speed adjuster
6. Part Power Trim Stop
7. Attachment Screw
8. Nut
9. Maximum Governing Stop Screw
10. Nut
11. Minimum Fuel Flow Adjuster
12. Nut

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 507 Fuel Control Unit Adjustments



PT6A-27 and PT6A-28

icn-00198-g000021232-001-01

1. Acceleration Adjuster Dome
2. Minimum Governing Speed Adjuster
3. Maximum Ng Speed Adjuster
4. GROUND-IDLE Deadband Adjuster
5. Rod End Clevis, Rear
6. Fuel Bypass Bleed to Tank
7. Interconnect Rod - Starting Control
8. Part Power Trim Stop (Set Block)
9. Starting Control FLIGHT-IDLE Max Stop
10. Locknut
11. Attachment Screw (Part Power Trim Stop)
12. Plug
13. Minimum Fuel Flow Adjuster
14. Screw
15. Nut

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

(1) Start engine (Ref. Para. 4. and 7.).

(2) Fuel Condition Lever - FLIGHT-IDLE (PT6A-21); Starting Control Lever - FLIGHT-IDLE (PT6A-27 and PT6A-28).

(3) Check that FLIGHT-IDLE linkage is on FLIGHT-IDLE stop (5, Fig. 506, PT6A-21), or (9, Fig. 507, PT6A-27 and PT6A-28).

WARNING: SHUT DOWN ENGINE BEFORE MAKING ANY ADJUSTMENTS ON THE ENGINE MOUNTED EQUIPMENT. THIS IS ESSENTIAL DUE TO CLOSE PROXIMITY OF PROPELLER AND EXHAUST NOZZLES.

CAUTION: INDIVIDUAL AIRFRAME MANUFACTURER'S MAY USE NG PERCENTAGES OF DIFFERENT VALUES TO THOSE STATED BELOW. REFER TO AIRCRAFT MAINTENANCE MANUAL.

(4) Check that Ng is 69% to 71%. If not, adjust stop (5 or 9) as applicable, until requirement is met.

NOTE: 1. It is permissible to adjust nut (4, Fig. 508) for certain installation (Ref. Aircraft Maintenance Manual).

NOTE: 2. Do this procedure after you do the rigging (Ref. AMM). If you do not get the necessary hi-idle speed or flight-idle speed with the interconnecting rods, adjust to the shortest length possible. Adjust the length as follows:

- Loosen the nut (15).
- Set the screw (14) inward with the hexagon key to decrease the travel of the lever to get the high-idle speed as necessary (Ref. AMM).
- Torque the nut (15) 10 to 12 lbf.in.

C. Gas Generator Speed (Ng) Adjustment

CAUTION: DURING ADJUSTMENTS AND NORMAL OPERATION, MAKE SURE THAT ENGINE OPERATING LIMITS ARE NOT EXCEEDED (REF. TABLES 503, 504 AND 505).

CAUTION: AT CERTAIN AMBIENT CONDITIONS, PART POWER TRIM SPEED WILL NOT BE ACHIEVED WITHOUT EXCEEDING ENGINE LIMITATIONS. IN THESE CONDITIONS, MAKE SURE FCU MAXIMUM SPEED SETTING IS MAINTAINED AS BENCH SET.

NOTE: Fuel control units are calibrated for maximum speed during benchcheck (overhaul). On PT6A-21 engines, the maximum Ng speed set ting should not be adjusted unless required. On PT6A-27 and PT6A-28 engines, the maximum Ng speed should be adjusted after engine installation or component replacement. The part power trim stop is a movable spacer placed between the maximum Ng stop and the power lever anvil, and represents a 1,700 rpm Ng speed decrement. After adjustments have been completed, the part power trim stop is returned to the stowed position on the FCU.

(1) For PT6A-21 Engines (Ref. Fig. 506):

(a) Do prestart check (Ref. Para. 4.).

(b) Position part power trim stop (6) so as to limit power control lever travel.

(c) Do a satisfactory start (Ref. Para. 7.).

(d) Advance power control lever to part power trim stop. Allow engine to stabilize for minimum of two minutes.

(e) Check that maximum gas generator speed (Ng) is 97.1% and adjust gas generator speed stop (9) as required.

(f) Shut down engine (Ref. Paras. 8. and 9.).

(g) Stow trim stop (6) and lockwire attachment screw (7).

(h) Lockwire maximum speed stop screw and locknut (9 and 8).

(2) For PT6A-27 and PT6A-28 Engines (Ref. Fig. 507):

CAUTION: THE MAXIMUM GOVERNING SPEED STOP IS PRESET ON ALL FUEL CONTROLS AND THE FOLLOWING PROCEDURE SHOULD NOT BE DONE UNLESS ABSOLUTELY NECESSARY. THIS PROCEDURE WILL ONLY BE POSSIBLE AT RELATIVELY HIGH AMBIENT TEMPERATURES.

CAUTION: DURING CHECKS AND ADJUSTMENTS, IT MAY BE NECESSARY TO EXCEED 97.1 % TO ACHIEVE MAXIMUM Ng AT ALTITUDE (101.5 % Ng).

(a) If FCU has been replaced, or fuel system lines disturbed, carry out a wet motoring run, followed immediately by a dry motoring run, to check fuel system for leakage.

(b) Unlock and slacken screw (11, Fig. 507). Move part power trim stop (8) to "in-use" position and secure by torquing screw (11) 25 to 30 lb.in.

(c) Start engine and warm up to operating temperature.

(d) Gradually advance power control lever as far as part power trim stop (8). Allow engine to stabilize at this setting for approximately five minutes with propeller in fine pitch.

NOTE: To reduce noise level, propeller pitch setting may be increased provided torque limit is not exceeded. Engine stabilization is essential to avoid erroneous instrument readings.

(e) Check that T5 limit is not exceeded, and that maximum Ng is 97.1 %. Adjust gas generator speed adjuster (3) as required, counterclockwise to increase speed, clockwise to decrease speed. One complete turn of adjuster changes Ng by approximately one percent.

(f) Shut down engine (Ref. Paras. 8. and 9.).

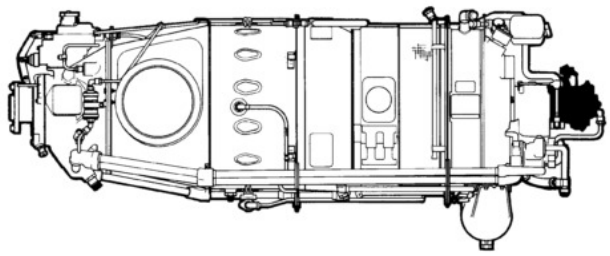
(g) Lockwire screw (3) and locknut (10).

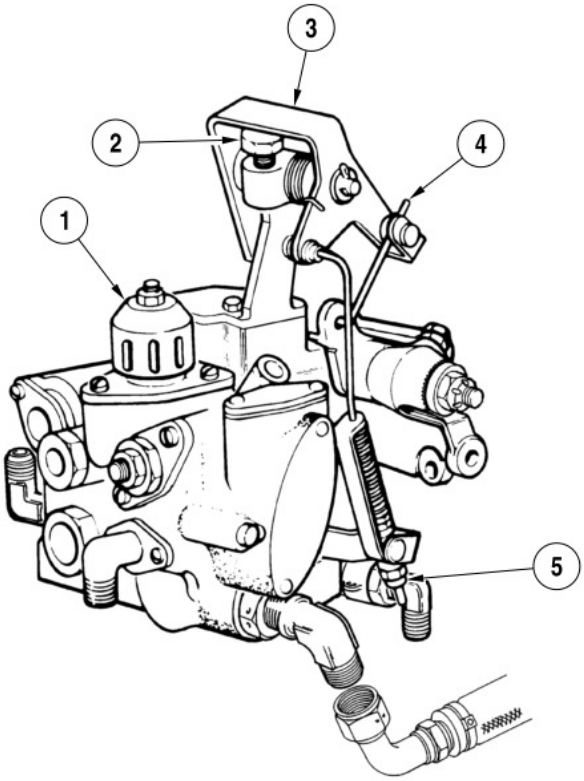
(h) Slacken screw (11), and return part power trim stop (9) to its "stowed" position. Tighten screw, torque 25 to 30 lb.in., and lockwire.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 508 Fuel Control Unit Adjustment (Engines with Flow Divider Valve)

(SHEET 1 OF 2)

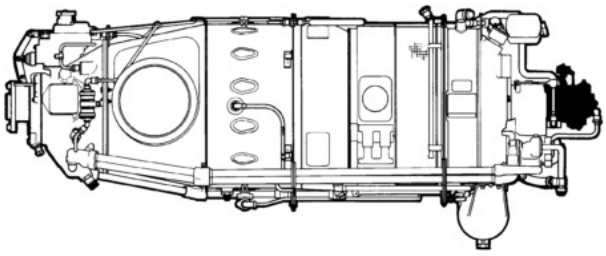


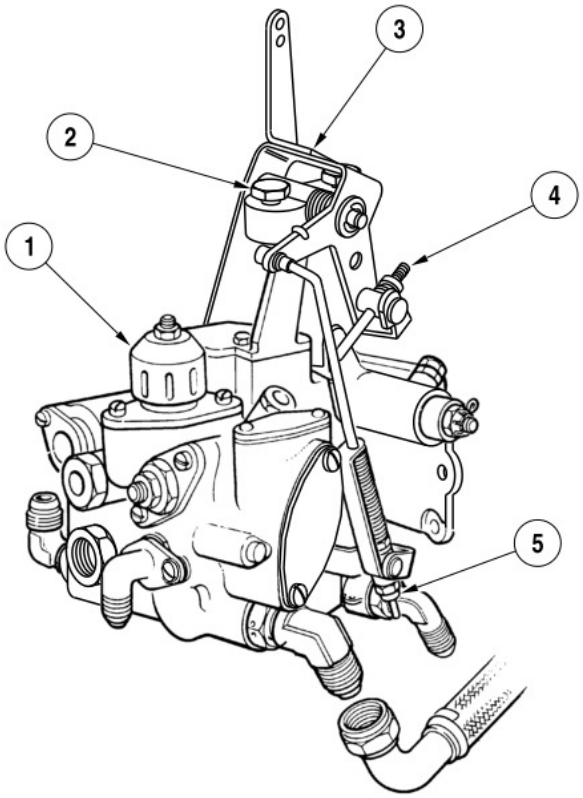


c71128

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

(SHEET 2 OF 2)





c80260

1. Acceleration Adjuster Dome
2. Flight-Idle Stop
3. Cut-off and Flight-Idle Linkage
4. Nut (ADJUSTABLE per Aircraft Maintenance Manual)
5. Nut (NOT ADJUSTABLE)

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

D. Manual Override System Static Check (PT6A-27)

(1) As part of the periodic inspection (Ref. 72-00-00, INSPECTION, Periodic Inspection), check the static operation of the manual override system, with the engine shutdown:

(a) Set cockpit Manual Override Control lever - OFF. Make sure the Manual Override lever on the FCU is in the OFF position (hard against its internal stop).

(b) Set cockpit Manual Override Control lever to the maximum forward position. Make sure the cockpit Manual Override Control lever does not bind and the Manual Override Lever on the FCU contacts the stop at the MAXIMUM FLOW position.

NOTE: Some resistance to movement of the cockpit Manual Override Control lever is normal.

(c) If either stop is not reached, check/adjust the airframe rigging (Ref. Aircraft Maintenance Manual).

E. Manual Override System Operational Check (PT6A-27)

CAUTION: OTHER THAN THE FOLLOWING CHECK, USE OF THE MANUAL OVERRIDE SYSTEM IS RESTRICTED FOR EMERGENCIES ONLY.

CAUTION: WHEN USING THE MANUAL OVERRIDE SYSTEM, THE ENGINE RESPONSE MAY BE MORE RAPID THAN WHEN USING THE POWER LEVER. A RAPID INCREASE IN MANUAL OVERRIDE LEVER POSITION TOWARDS MAXIMUM IS TO BE AVOIDED, AS THIS CAN CAUSE ENGINE SURGE, ENGINE OVER-TEMPERATURE, Ng OVER-SPEED OR OVER-TORQUE.

(1) Following an FCU or engine change (Ref. 71-00-00, ADJUSTMENT/TEST, Table 501) or as part of an annual airframe zone inspection (Ref. Aircraft Maintenance Manual), check the serviceability of the manual override system:

(a) Do the Manual Override System static check (Ref. Subpara. D.).

(b) Set Manual Override Control lever - OFF.

(c) Start the engine (Ref. Para. 4. and 7.).

(d) Fuel Condition lever - GROUND-IDLE.

(e) Power Control lever - GROUND-IDLE.

(f) Slowly advance Manual Override Control lever observing ITT, torque and Ng at all times. Keep Ng maximum increase under 4% per second.

NOTE: Some movement of the Manual Override Control lever, resulting in no increase in engine power, is normal.

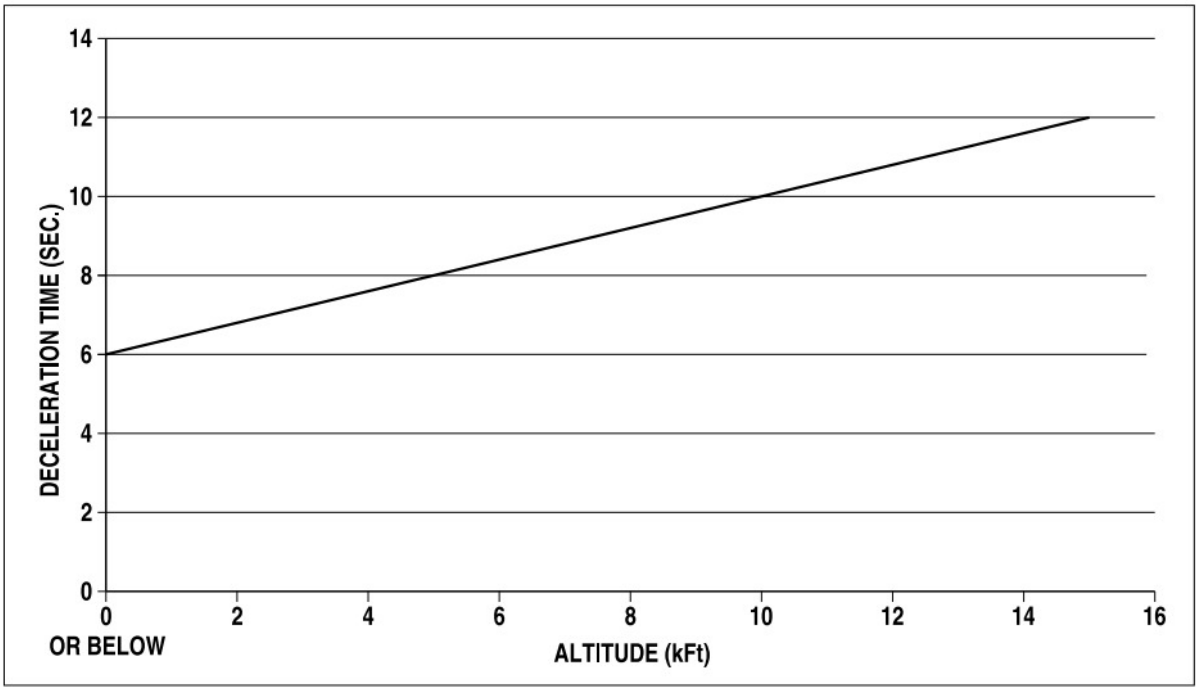
(g) Slowly increase Ng until a 15% increase above ground-idle speed is obtained. This indicates proper functioning of the Manual Override System.

(h) Slowly reduce the Manual Override Control lever to OFF, keeping Ng decrease less than 4% Ng per second.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

F. Deceleration Check (Ref. Fig. 509)

Figure 509 Deceleration Check



c228856

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

(1) Do a check for ground idle speed (Ref. Para. 11.A.) and record in logbook.

(2) With the engine still running, set to min. idle and let instruments stabilize for one minute minimum.

(3) If equipped, make sure that the cockpit Manual Override Control lever is to the OFF position.

(4) Set engine Ng speed to 85% and propeller speed to MAX and stabilize engine for ten seconds.

(5) Slam decel the power control lever in less than one second to ground idle speed.

(6) Note and record in logbook time to reach Ng of 60% or low idle speed, whichever comes first.

(7) If deceleration time is more than the limit, contact P&WC.

12. Oil Pressure Check

A. Procedure

(1) Start engine (Ref. Para. 4. and 7.).

(2) Run engine at convenient gas generator speed above 72% to stabilize oil temperature within operating range.

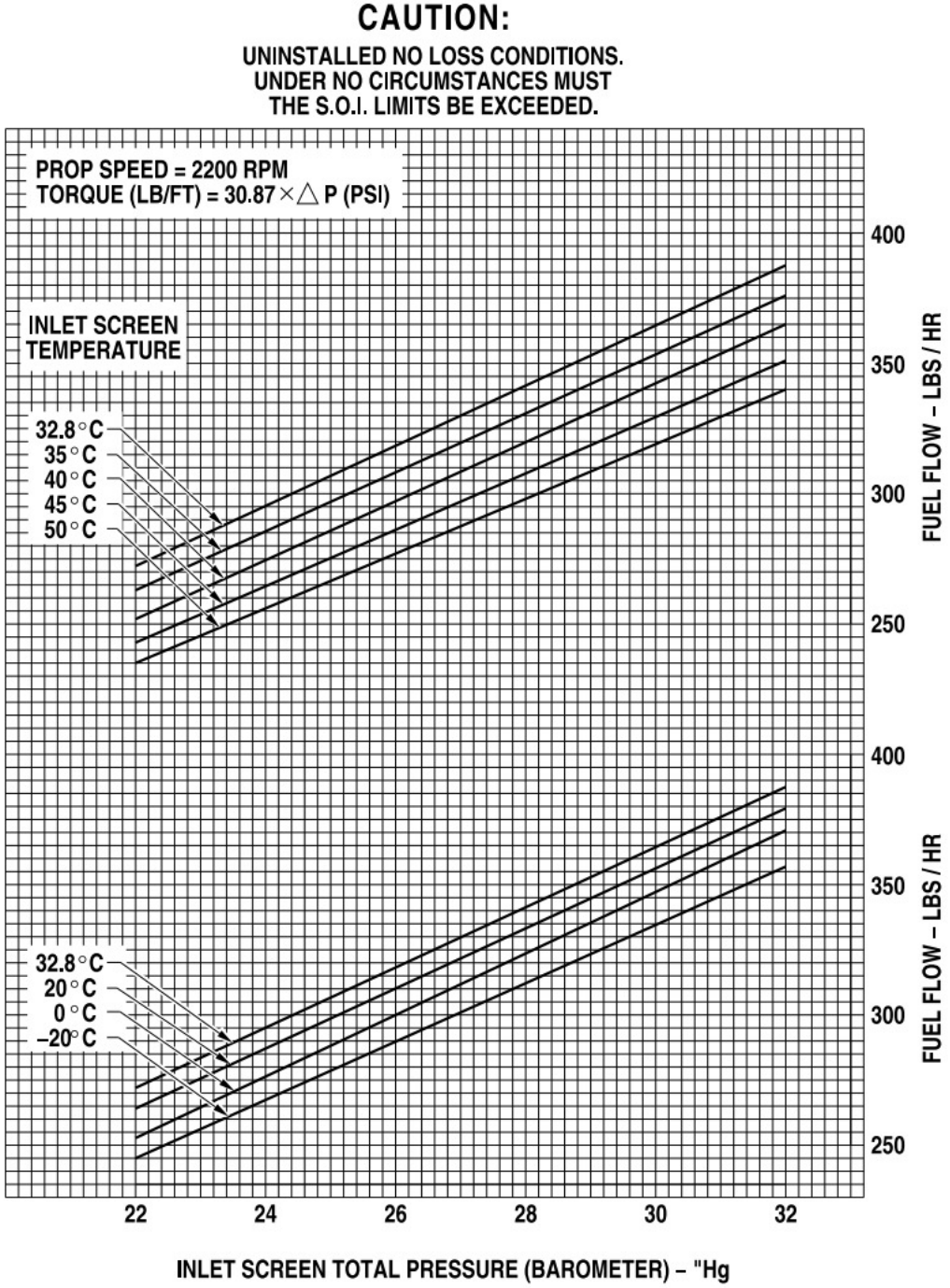
(3) After instrument readings have stabilized, check that oil pressure is within limits specified in relevant engine operating limit table (Ref. Tables 503, 504 and 505). If not, change quantity of spacers in oil pressure relief valve (Ref. 72-60-00).

13. Engine Performance Check (Ref. Figs. 510, 511 and 512)

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 510 Engine Power Assurance Check Curve

(SHEET 1 OF 2)

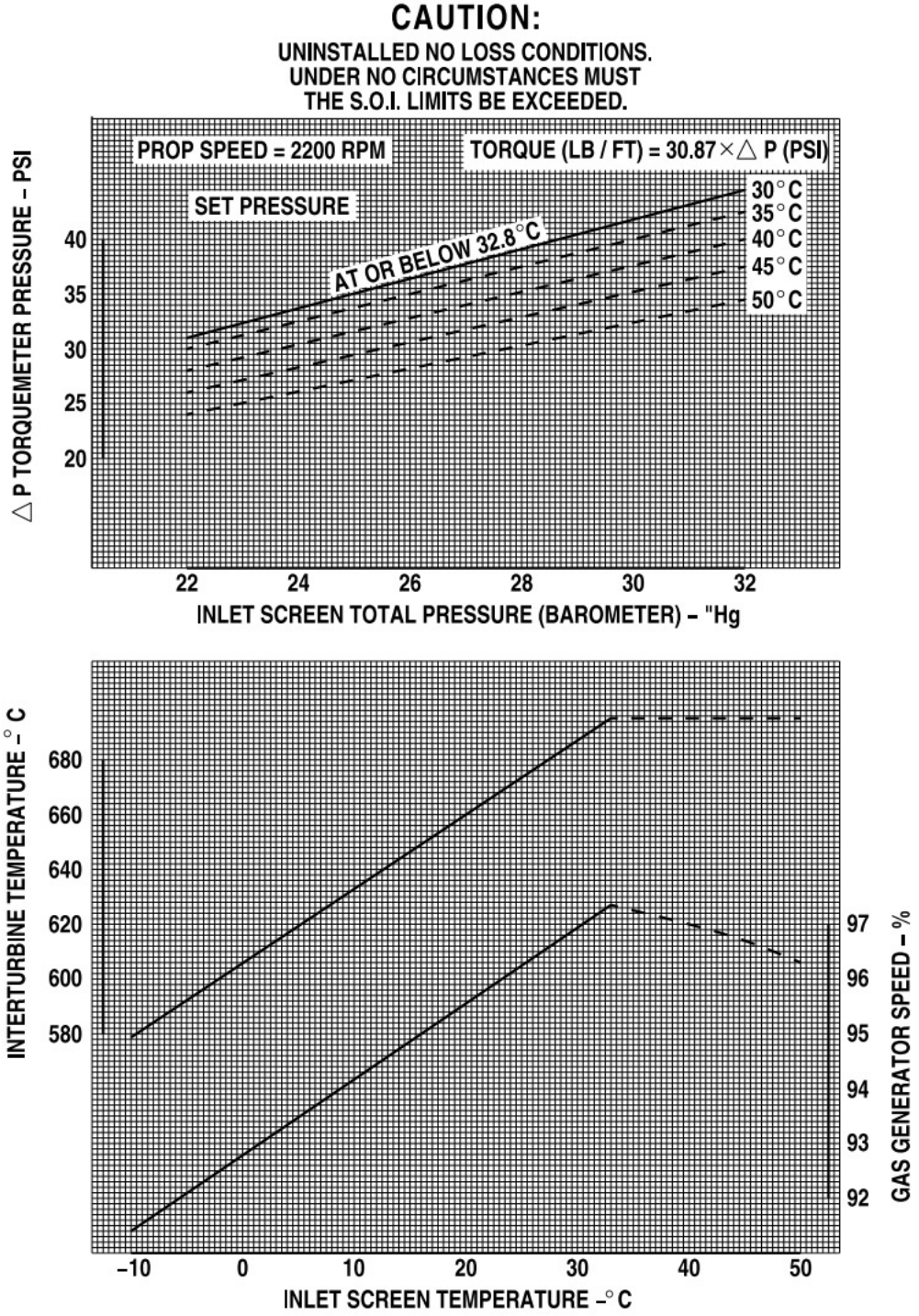


PT6A-21

c166610

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

(SHEET 2 OF 2)



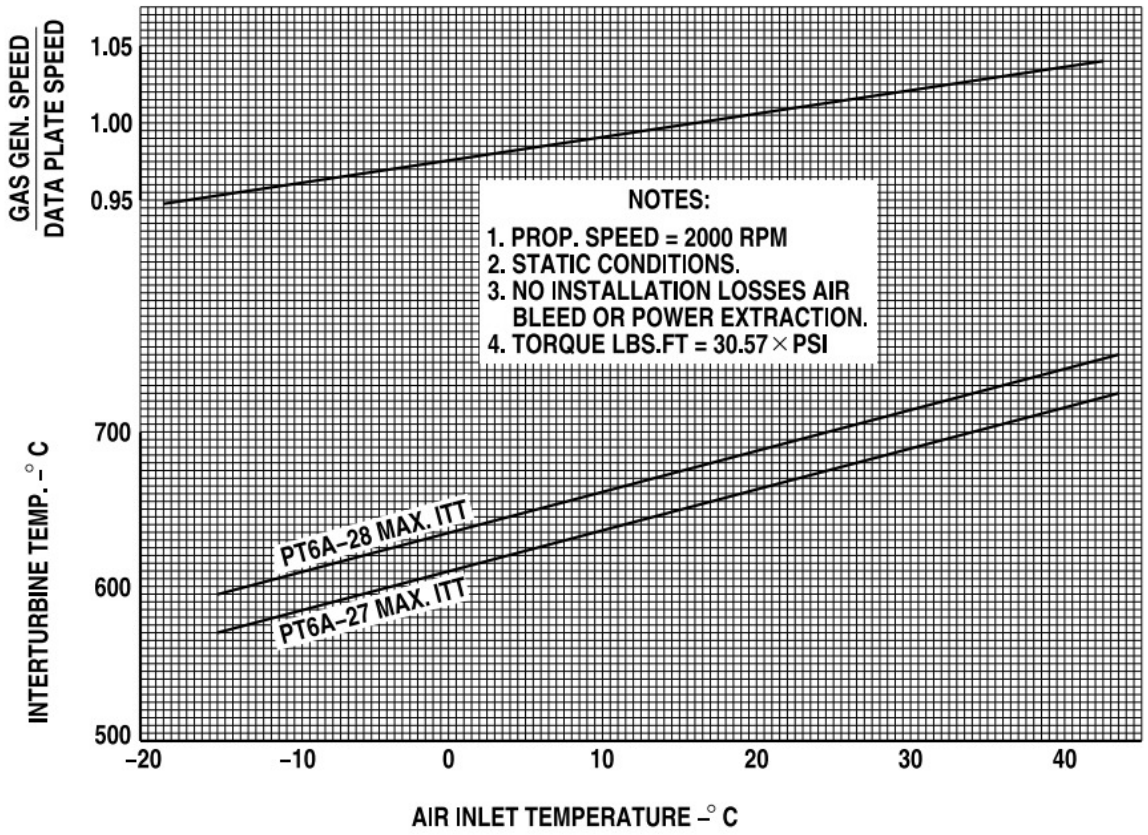
PT6A-21

c166609

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 511 Engine Performance and Data Plate Trim Speed Curve

(SHEET 1 OF 2)

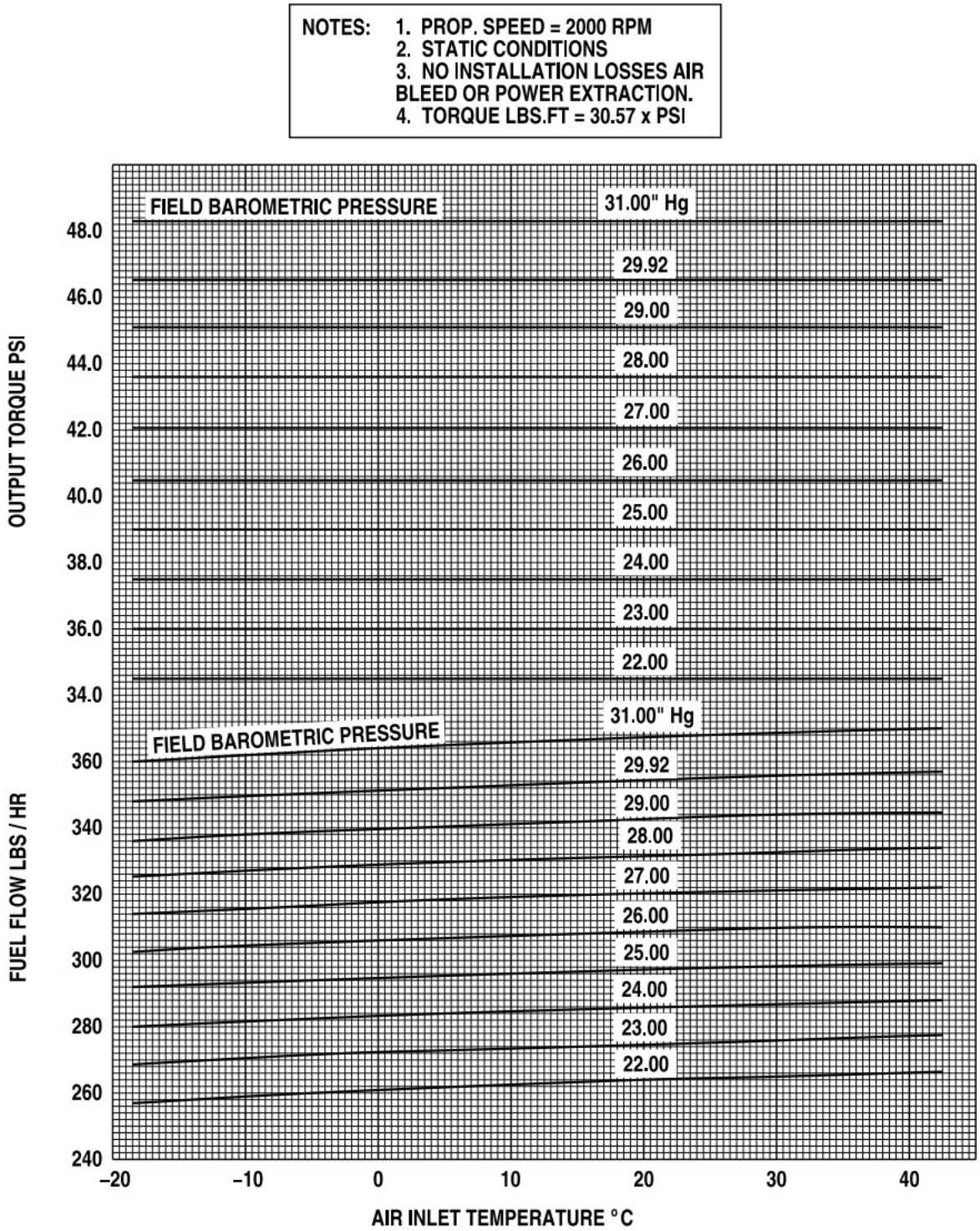


PT6A-27 and PT6A-28

c469d1

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

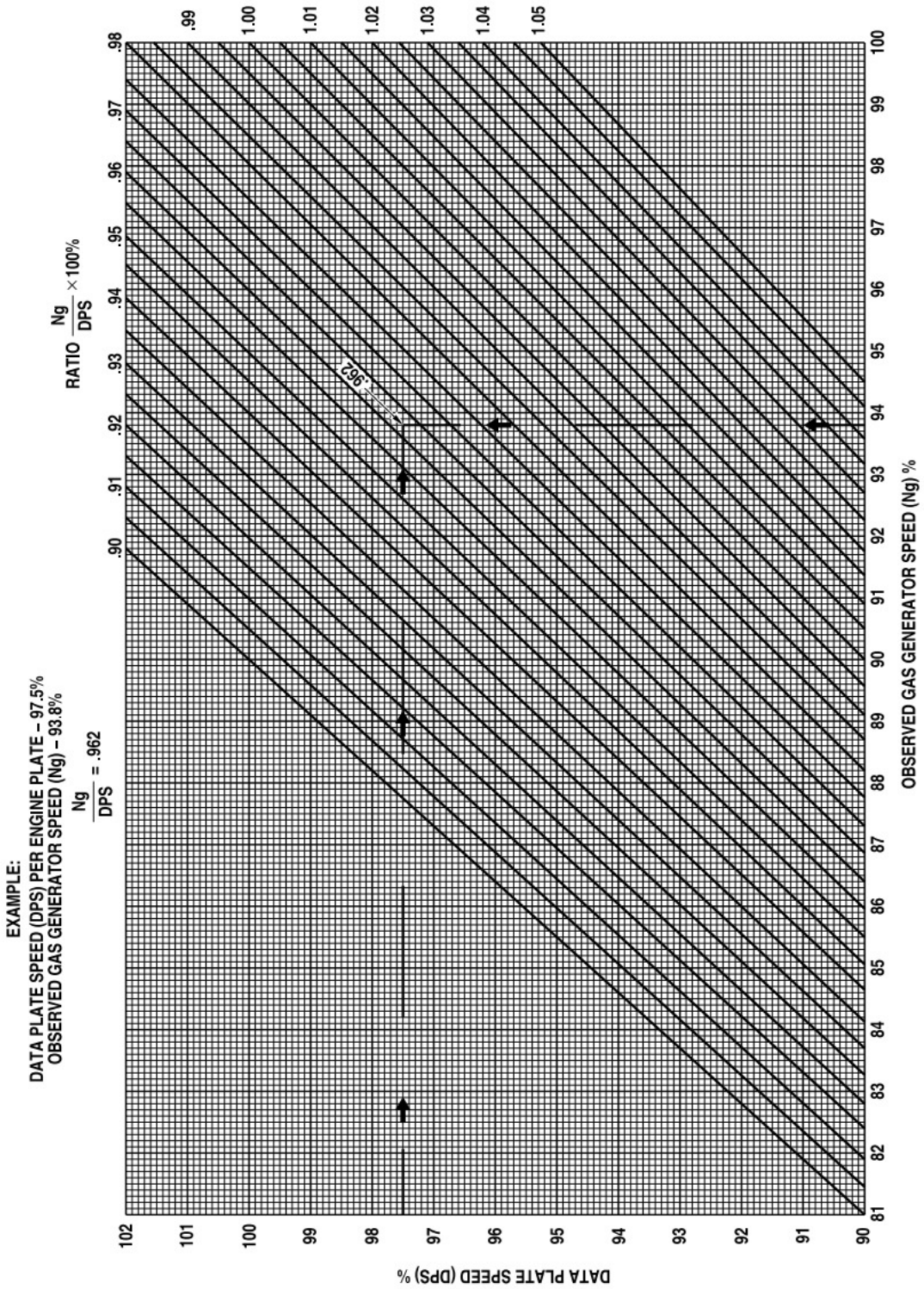
(SHEET 2 OF 2)



PT6A-27 and PT6A-28

c469d2

Figure 512 Data Plate Speed Ratio Curve



PT6A-27 and PT6A-28

c183126

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

A. General

Data plate trim speed is gas generator speed (Ng) obtained with engine running at reference power on a standard day (15 °C; 59 °F). This value is determined from final acceptance test results, and is stamped or engraved on the engine data plate in rpm, and as a percentage of maximum Ng at a reference power of 544 SHP/DELRTH.

The engine power assurance/performance (Ref. Figs. 510 and 511) and data plate trim speed check curves (Ref. Fig. 512) enable engine performance to be checked over a wide range of ambient temperatures without exceeding torque and ITT limits.

All forms of engine deterioration are accompanied by an increase in ITT and Wf at a given power as indicated by the torquemeter gage. Compressor deterioration, generally due to a build-up of dirt deposits on compressor blades, is usually accompanied by an increase in ITT, Wf and Ng. In this condition, performance may be recovered by compressor washing (Ref. CLEANING). Deterioration of hot section components will also be accompanied by an increase in ITT and Wf, and may be accompanied by a decrease in Ng.

B. Procedure

(1) Determine engine performance check parameters:

(a) Obtain ambient air temperature (Ta) in degrees Celsius..

(b) Obtain field barometric pressure (Pa) (not corrected to sea level) in inches of mercury.

(c) From engine performance check curves (Ref. Figs. 510 and 511), determine and record torque pressure, fuel flow (Wf), Ng/data plate trim speed ratio and interturbine temperature (ITT).

(2) For PT6A-21 engines, do engine performance check:

(a) Close P3 air supply to airframe auxiliaries.

(b) Start engine and run at GROUND-IDLE for five minutes to stabilize operating temperatures.

(c) Set Propeller Control Lever to 100% Np (2200 rpm) and Power Control Lever to obtain previously determined torque setting. Let engine instrumentation stabilize.

(d) Record observed Wf, Ng and ITT and compare with previously recorded figures.

(e) Make sure the values observed during engine performance check are within the following limits:

1 + 15 lb/hr fuel flow. If fuel flow is more than 75 lb/hr below chart value, check instrumentation.

2 Maximum gas generator speed is not exceeded.

3 Maximum interturbine temperature line is not exceeded. If temperature is more than 75°C below target temperature, check instrumentation.

(f) Values outside the limits specified in step (e), preceding, are not necessarily cause for engine rejection until all relevant trouble shooting techniques have been exhausted (Ref.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

72-00-00, FAULT ISOLATION). Replacement and/or repair of hot section parts can affect engine performance parameters. Any increase in speed should be recorded for future reference. The importance of monitoringtheperformance trend of an engine between overhauls, and during itslife, cannot be overemphasized.

(3) For PT6A-27 and PT6A-28 engines, do engine performance check:

(a) Close P3 air supply to airframe auxiliaries.

(b) Start engine and run at GROUND-IDLE for five minutes to stabilize operating temperatures.

(c) Set Propeller Control Lever to 91% Np and Power Control Lever to obtain previously determined torque setting. Let engine instrumentation stabilize.

(d) Record observed Wf, Ng and ITT and compare with previously recorded figures.

(e) Make sure the values observed during engine performance check are within the following limits:

1 + 15 lb/hr fuel flow. If fuel flow is more than 75 lb/hr below chart value, check instrumentation.

2 ± 0.02 Ng/data plate trim speed ratio observed (gas generator speed/data plate trim speed). Refer to Fig. 512 to compute ratio.

3 ITT: Below maximum limits. If temperature is more than 75°C (167°F) below target temperature, check instrumentation

(f) Values outside the limits specified in step (e), preceding, are not necessarily cause for engine rejection until all relevant trouble shooting techniques have been exhausted (Ref. 72-00-00, FAULT ISOLATION). Replacement and/or repair of hot section parts can affect engine performance parameters. Any increase in speed should be recorded for future reference. The importance of monitoringtheperformance trend of an engine between overhauls, and during itslife, cannot be overemphasized.

14. Acceleration Adjustments

A. Procedure

(1) Start engine and operate at GROUND-IDLE for five minutes to allow stabilization of temperatures.

(2) Slowly increase power to maximum (or max. allowable on ground; Ref. Aircraft Maintenance Manual). Note and record Ng. Mark position of power control lever on cockpit console.

(3) Reduce power and set Ng to High Idle. If high idle value not stated in AMM, set Ng to 64%.

(4) Rapidly (in less than 1.0 second), move power control lever to position marked in step (2).

(5) Note and record time required for Ng to reach 93.5% of the value recorded in step (2). Record ambient temperature.

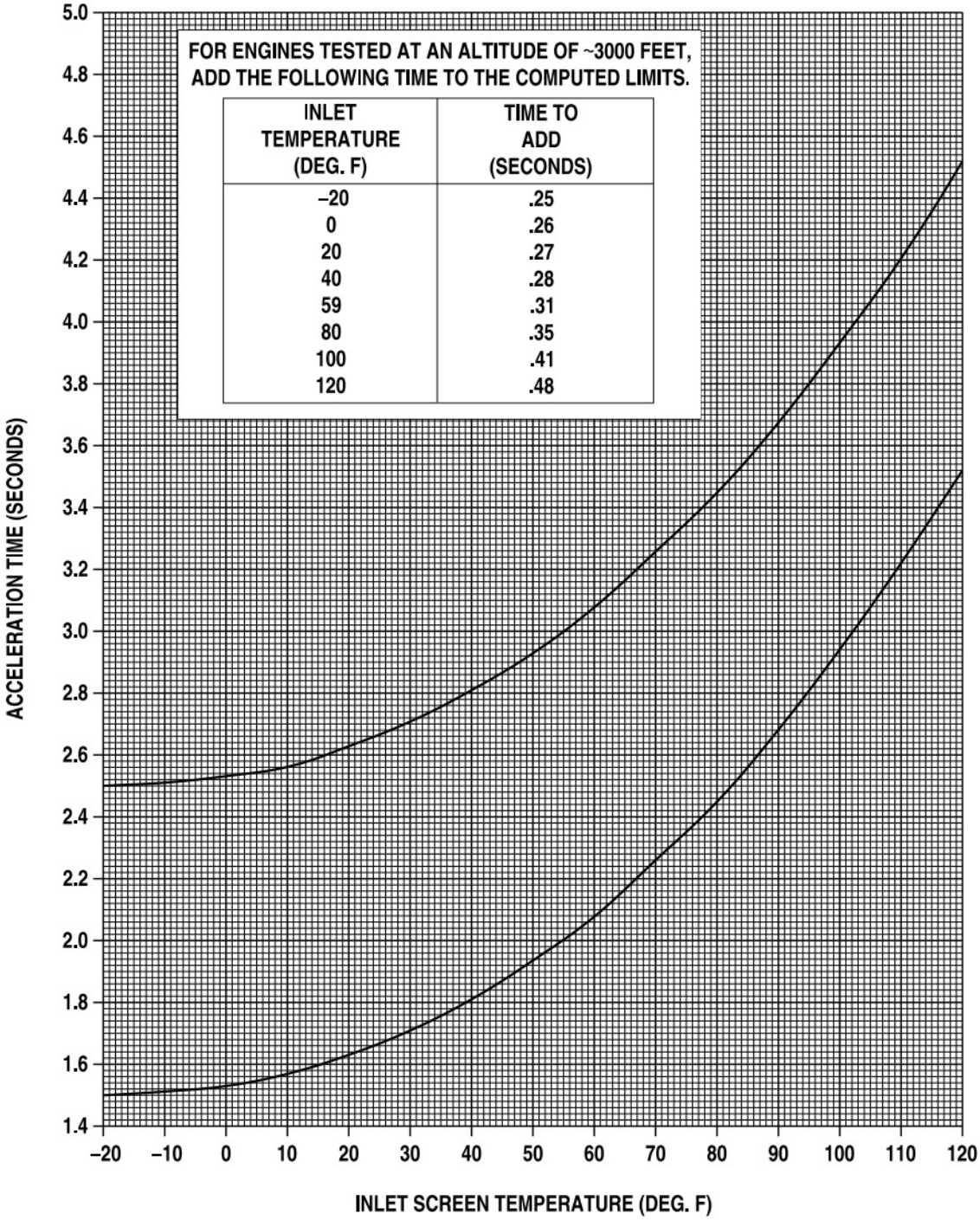
(6) Acceleration time must fall within the range specified in figure 513. If not, rotate acceleration adjustment dome (1, Figs. 506 and 507) one click at a time until requirement is met. Rotate dome clockwise to reduce acceleration time (i.e. faster acceleration) or

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

counterclockwise to increase acceleration time (slower acceleration). If requirement cannot be met within a maximum of three clicks, clockwise or counterclockwise, replace fuel control unit.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 513 Acceleration Time Vs. Ambient Temperature



c36753

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

15. Torquemeter Function Test

A. Procedure

(1) Determine and record field barometric pressure (pressure altitude (ft)).

NOTE: Pressure altitude (PA) can be obtained by contacting local flight service for uncorrected pressure reading, or by setting altimeter to 29.92 Hg. Position of needle indicates pressure altitude in feet. Do not use sea level pressure.

(2) Determine and record indicated outside air temperature (IOAT) in degrees Celsius.

NOTE: To make sure result of torquemeter function test is accurate, determine and record IOAT immediately before engine start. Use IOAT value recorded to determine take-off torque (Ref. Step (3)).

(3) Use the flight manual chart to determine the take-off torque (target torque) at which to set the engine using pressure altitude (ft) and IOAT recorded at step (1) and step (2) respectively. Record the value.

(4) Start the engine.

(5) Switch off the generator, bleed air and the air conditioning.

(6) Check propeller control lever is in high rpm.

(7) Check oil temperature is above 60 °C.

(8) Set Ng 500 rpm above GROUND-IDLE (Ref. Operating Limitations).

(9) Record Ng, Np and engine torque (Tq).

(10) Accelerate engine slowly and uniformly at a rate of 2000 rpm per second until target torque recorded at step (3) is reached.

(11) Allow engine to stabilize for two minutes.

(12) Reduce Ng slowly and uniformly at a rate of 2000 rpm per second to Ng set at Step (8) .

(13) Record Ng, Np and Tq.

(14) Repeat steps (10) to (13) .

(15) Reduce Ng to GROUND-IDLE.

(16) Shutdown engine (Ref. Para. 8.).

(17) The engine torque indication system functions correctly when the difference between Tq recorded in steps (9) , (13) and (14) is equal to or less than 200 lb.ft.

(18) If the difference between the torque readings is in excess of 150 lb.ft. , check the aircraft indication system. If the aircraft indicating system check is satisfactory, the power section is recommended to be removed and the torquemeter piston, cylinder and servo valve inspected at an authorized facility for excessive wear, scoring, fretting, etc.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement